Диффуры. Бадма Александрович

ПЗ 1

Решение - семейство функций

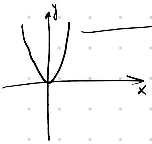
$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

$$y = y(x, C_1, \dots, C_n)$$

$$y' = f(x,y) = \operatorname{tg}\alpha$$

Метод изоклин

$f(x,y) = k = const$ – изоклина



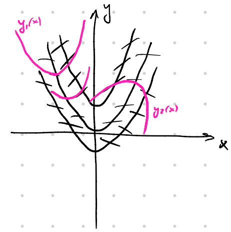
1) $y' = y - x^2$

$y - x^2 = k$

$y = x^2 + k$

k	α	
0	0	$y = x^2$
1	$\frac{\pi}{4}$	$y = x^2+1$
-1	$-\frac{\pi}{4}$	$y = x^2-1$

● - решения

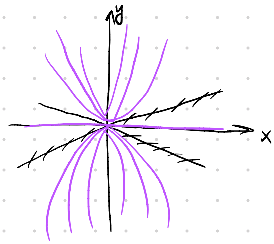


6) $xy' = 2y$

$xk = 2y$

$y = \frac{xk}{2}$

k	α	
0	0	$y=0$
1	$\frac{\pi}{4}$	$y = \frac{x}{2}$
-1	$-\frac{\pi}{4}$	$y = -\frac{x}{2}$

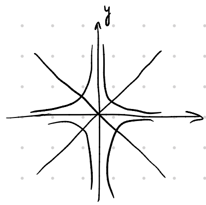


7) $xy' + y = 0$

$xk + y = 0$

$y = -xk$

k	α	
0	0	$y=0$
1	$\frac{\pi}{4}$	$y=-x$
-1	$-\frac{\pi}{4}$	$y=x$



$$y = y(x, C_1, \dots, C_n)$$

$$\begin{cases} y' = \\ \vdots \\ y^{(n)} = \end{cases}$$

17) $y = ax^2 + be^x$

$$\begin{cases} y' = 2ax + be^x \\ y'' = 2a + be^x \end{cases}$$

$y' - y'' = 2ax - 2a$

$\frac{y'-y''}{2(x-1)} = a$

$y'' = \frac{y'-y''}{x-1} + be^x$

$b = \frac{y''(x-1) - y' + y''}{(x-1)e^x}$

$y = \frac{y'-y''}{2(x-1)} x^2 + \frac{y''(x-1)-y'+y''}{x-1}$

Уравнение с раздел. переменными

$y' = f(x)g(y)$ / $M(x)N(y)dx + P(x)Q(y)dy = 0$

Проверка: $g(y) = 0$

$\frac{dy}{dx} = f(x)g(y)$

$\int \frac{dy}{g(y)} = \int f(x)dx$

$G(y) + C_1 = F(x) + C_2$

$G(y) = F(x) + C$

51) $xydx + (x+1)dy = 0$

$xydx = -(x+1)dy$

$-\frac{xdx}{x+1} = \frac{dy}{y}$

$-\int \frac{x}{x+1}dx = \ln|y| + C_2$

$-\int 1 - \frac{1}{x+1} dx = \ln|y| + C_2$

$-x + \ln|x+1| + C_1 = \ln|y| + C_2$

$y = e^{\ln|x+1| - x + C}$

Проверка: $x+1=0$

$\boxed{x=-1}$ – реш.

$\boxed{y=0}$

54) $y'\operatorname{ctg}x + y = 2$

Нач. усл.: $y(x) \to -1$, $x \to 0$

$\frac{dy}{dx}\operatorname{ctg}x = 2 - y$

$dy \operatorname{ctg}x = (2-y)dx$

$\int \frac{dy}{2-y} = \int \operatorname{tg}x dx$

$-\ln|2-y| = -\ln|\cos x| + C$

$2 - y = C_0 \cdot \cos x$

$y = -C_0\cos x + 2$

Проверка:

$\cancel{\boxed{y=2}}$

$y(x) \to -1$, $x \to 0$

$C_0 = 3$

$y = -3\cos x + 2$

Задача Коши

$$y' = f(x, y)$$
$$y(x_0) = y_0$$

(56) $xy' + y = y^2$

$y(1) = 0{,}5$

$$\frac{dx}{dy} x + y = y^2$$
$$\frac{dx}{dy} x = y^2 - y$$
$$x dy = (y^2 - y) dx$$
$$\frac{dy}{y^2 - y} = \frac{dx}{x}$$

$$\int \frac{dy}{(y-\frac{1}{2})^2 - \frac{1}{4}} = \int \frac{dx}{x}$$

(63) $y' - y = 2x - 3$

$$\frac{dy}{dx} = y + 2x - 3$$
$$dy = (y + 2x - 3) dx$$
$$dy = y dx + 2x dx - 3dx$$
$$y + 2x - 3 = t$$
$$dt = dy + 2dx$$
$$dy = dt - 2dx$$

$$dt - 2dx = t dx$$
$$dt = (t+2) dx$$
$$\frac{dt}{t+2} = dx$$
$$\int \frac{dt}{t+2} = \int dx$$
$$\ln|t+2| = x + C$$
$$t + 2 = e^x \cdot C$$
$$t = e^x \cdot C - 2$$
$$y + 2x - 3 = e^x \cdot C - 2$$

$$y = e^x \cdot C - 2x + 1$$

Проверка:

$t = -2$

Однородные уравнения

$$f(x, y): \Rightarrow f(tx, ty) = t^n f(x, y)$$

$y' = f\left(\frac{y}{x}\right)$ или $M(x,y)dx + N(x,y)dy = 0$ замена $y = t(x) \cdot x$

$$(\overset{M}{y^2 - 2xy}) dx + \overset{N}{x^2} dy = 0$$
$$M(tx, ty) = t^2 y^2 - 2txty = t^2(y^2 - 2xy) = t^2 M$$
$$N(tx, ty) = t^2 x^2 = t^2 N$$
$$y = tx$$
$$dy = x dt + t dx$$
$$(t^2 x^2 - 2x^2 t) dx + x^2 (x dt + t dx) = 0$$
$$dx(t^2 x^2 - x^2 t = -x^2 dt$$
$$\int -\frac{1}{x} dx \int \frac{1}{t(t-1)} dt$$
$$c - \ln|x| = \ln\left|\frac{t-1}{t}\right|$$
$$c - \ln|x| = \ln\left|\frac{y/x - 1}{y/x}\right|$$

$x = 0$ ✓

$t = 0$ ✓

$t = 1$ ✓

$y = 0$ ✓

$$xy' = y \cos \ln \frac{y}{x}$$
$$y = tx$$
$$y' = \frac{dy}{dx}$$
$$dy = t dx + x dt$$
$$y' = \frac{t dx + x dt}{dx}$$

$$xy' = tx \cos \ln t$$
$$x \frac{t dx + x dt}{dx} = tx \cos(\ln t)$$
$$\frac{t dx + x dt}{t \cos(\ln t)} = \frac{\cancel{x} dx}{\cancel{x}}$$
$$\int \frac{d(\ln t)}{(\cos(\ln t) - 1)} = \int \frac{dx}{x}$$

$$-\int \frac{d(\ln t)}{(1 - \cos(\ln t))} = \int \frac{dx}{x}$$
$$-\int \frac{d \ln t/2}{\sin^2\left(\frac{\ln t}{2}\right)} = \ln|x| + C$$

$$y' = f\left(\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}\right)$$

$$y' = 2\left(\frac{y+2}{x+y-1}\right)^2$$

$y = v+\alpha$

$\alpha = -2$

$y = v-2$

$x = u+\beta$

u

$u+\beta+v-2=1$

$\beta-2=1$

$\beta=3$

$$\frac{dv}{du} = 2\left(\frac{v}{u+v}\right)^2$$

$v = t\cdot u$

$dv = u\,dt + t\,du$

$$u\,dt + t\,du = 2\left(\frac{tu}{u(t+1)}\right)^2$$

$$u\,dt = \left(2\left(\frac{t}{t+1}\right)^2 - t\right)du$$

Линейные ур-я 1го порядка

$$y' + p(x)y = q(x)$$

1. Метод вариации постоянной

$q(x) = 0$ – одноp.

$q(x) \neq 0$ – неодноp.

$y_{\text{общ. неодн.}} = y_{\text{одн}} + y_{\text{част}}$

$y' + py = 0$

$\Downarrow$

$y = y(c,x)$

$y = y(C(x),x)$

$\Downarrow$

$y + py = q$

$C(x) \longrightarrow$ ур-е с раздел. перем. относительно C

$C(x) = C(x, C_0)$

$y = y(C(x,C_0),x)$ – ответ

138. $y' + y\,tg\,x = sec\,x$

$y' + y\,tg\,x = 0$

$\frac{dy}{dx} + y\,tg\,x = 0$

$-dy = y\,tg\,x\,dx$

$-\frac{dy}{y} = tg\,x\,dx$

$\ln(y) = \ln(\cos x)$

$y = C\cdot\cos x$

$C'(x)\cos x - \cancel{\sin x\, C(x)} + \cancel{C\cos x\, tg\,x} = \frac{1}{\cos x}$

$C'(x)\cos x = \frac{1}{\cos x}$

$dC(x)\cos x = \frac{dx}{\cos x}$

$C(x) = \int\frac{dx}{\cos^2 x}$

$C(x) = tg(x) + C$

$y = (tg\,x + c)\cos x = \sin x + C\cdot\cos x$

2. Метод Бернулли

$y = uv \Rightarrow u'v + v'u + puv = q$

$$\begin{cases} u'\cancel{v} + pu\cancel{v} = 0 \to u(x)! \\ uv' = q \end{cases}$$

$y' + y\,tg\,x = sec\,x$

$y = uv$

$u'v + v'u + uv\,tg\,x = \frac{1}{\cos x}$

$$\begin{cases} u'v + uv\,tg\,x = 0 \\ v'u = \frac{1}{\cos x} \end{cases}$$

$u' + u\,tg\,x = 0$

$\frac{du}{dx} = -u\,tg\,x$

$-\frac{du}{u} = \frac{\sin x\,dx}{\cos x}$

$u = \cos x$

$v'\cos x = \frac{1}{\cos x}$

$\frac{dv}{dx} = \frac{1}{\cos^2 x}$

$dv = \frac{dx}{\cos^2 x}$

$y' = y^4\cos x + y\,tg\,x$

$y' - y\,tg\,x = y^4\cos x \quad |:y^4$

$\frac{y'}{y^4} - \frac{tg\,x}{y^3} = \cos x$

$z = \frac{1}{y^3} = y^{-3} \quad z' = -3y^{-4}\cdot y' = 3\frac{y'}{y^4}$

$\frac{z'}{-3} - z\,tg\,x = \cos x$

$z' + 3z\,tg\,x = -3\cos x$

$z' + 3z\,tg\,x = 0$

$-\frac{dz}{z} = -3\,tg\,x\,dx$

$\ln z = 3\ln\cos x + \ln c$

$z = \cos^3 x$

$C\,3\cos^2 x(-\sin x) + C'\cos^3 x + 3C\cos^3 x\,tg\,x = -3\cos x$

$C'\cos^{3\,2} x = -3\cos x$

$C = -3\,tg\,x + C_0$

$z = (-3\,tg\,x + C_0)\cos^3 x$

$$y = \frac{1}{\sqrt[3]{(-3\,tg\,x + C_0)\cos^3 x}}$$

$xy' - (2x+1)y + y^2 = -x^2$

$y' - \frac{2x+1}{x} y + \frac{1}{x} y^2 = -x$

$y_0 = ax + b$

$ax - (2x+1)(ax+b) + (ax+b)^2 = -x^2$

$-2a + a^2 = -1$

$a = 1$

$a - 2b - a + 2ab = 0$

$-b + b^2 = 0$

$\left[\begin{array}{l} b=0 \\ b=1 \end{array}\right.$

$\underline{y_0 = x}$

$y = x + z$

$x(1+z') - (2x+1)(x+z) + (x+z)^2 = -x^2$

$\cancel{x} + xz' - \cancel{2x^2} - \cancel{x} - \cancel{2xz} - z + \cancel{x^2} + \cancel{2xz} + z^2 = -\cancel{x^2}$

$xz' - z + z^2 = 0$

$z' - \frac{1}{x} z = -\frac{1}{x} z^2$

$\int_0^x (x-t) y(t) dt = 2x + \int_0^x y(t) dt$

Уравнение в полных диф-х

$M dx + N dy = 0$

$\exists\, U : dU = M dx + N dy = U_x' dx + U_y' dy$

$dU = 0$

$U = const$

$U''_{xy} = U''_{yx}$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

① $U_x' = M$

$U = \int M dx = M_1 + \varphi(y)$

$U_y' = N$

$\frac{\partial M_1}{\partial y} + \varphi' = N \Rightarrow \varphi(y)$

$U = C$

② $U = \int_{(x_0,y_0)}^{(x,y)} M dx + N dy =$

$U(x,y) - U(x_0,y_0)$

$= \int_{x_0}^{x} M(x, y_0) dx + \int_{y_0}^{y} N(x,y)\, dy$

③ заметим, что очевидно — ответ

диф-л — линейный оператор

189

$\frac{y}{x} dx + (y^3 + \ln x) dy = 0$

③ $\frac{y}{x} dx + \ln x\, dy + y^3 dy = 0$

$d(y \ln x) + d\left(\frac{y^4}{4}\right) = 0$

$d\left(y \ln x + \frac{y^4}{4}\right) = 0$

$y \ln x + \frac{y^4}{4} = C$

① $U = U_x' dx + U_y' dy$

$U_x' = \frac{y}{x}$

$U = \int \frac{y}{x} dx = y \ln|x| + \varphi(y)$

$(y \ln|x| + \varphi(y))_y' = y^3 + \ln x$

$\varphi'(y) = y^3$

$\varphi(y) = \frac{y^4}{4}$

$U = y \ln|x| + \frac{y^4}{4} = C$

② $\int_1^x 0dt + \int_0^y (y^3+\ln x)dy = 0 + \frac{y^4}{4}\Big|_0^y + y\ln x\Big|_0^y = \frac{y^4}{4} + y\ln x$

$u = \frac{y^4}{4} + y\ln x = C$

Инт. множитель

$Mdx + Ndy = 0 \mid \cdot \mu(x,y)$

$\mu M dx + \mu N dy = 0$

$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}$

① а) $\mu = \mu(x)$

$\mu\frac{\partial M}{\partial y} = \mu' N + \mu\frac{\partial N}{\partial x}$

$\mu\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \mu' N$

$(\ln\mu)' = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$

б) $\mu = \mu(y)$

$(\ln\mu)' = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$

195 $(x^2+y^2+x)dx + ydy = 0$

$(\ln\mu') = \frac{2y}{y} = 2$

$\ln\mu = 2x$

$\mu = e^{2x}$

$e^{2x}(x^2+y^2+x)dx + e^{2x}ydy = 0$

$(x^2+x)e^{x}dx + y^2e^{2x}dx + e^{2x}y\,dy = 0$

$d/\int(x^2+x)e^{2x}dx + d\left(y^2\frac{e^{2x}}{2}\right) = 0$

$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}$

$\mu = \mu(\omega),\ \omega = \omega(x,y)$

$\mu'\frac{\partial\omega}{\partial y}M + \mu\frac{\partial M}{\partial y} = \mu'\frac{\partial\omega}{\partial x}N + \mu\frac{\partial N}{\partial x}$

211 $(2x^2y^2+y)dx + (x^3y-x)dx = 0 \mid \cdot\mu$

$\mu'\frac{\partial\omega}{\partial y}(2x^2y^2+y) + \mu(4x^2y+1) = \mu'\frac{\partial\omega}{\partial x}(x^3y-x) + \mu(3x^2y-1)$

$\omega = xy$

$\mu'(2x^3y^2+xy) + \mu(4x^2y+1) = \mu'(x^3y^2-xy) + \mu(3x^2y-1)$

$\mu'(x^3y^2+2xy) + \mu(x^2y+2) = 0$

$\frac{d\mu}{d\omega} = -\frac{\mu}{\omega}$

$\mu = \frac{1}{\omega} = \frac{1}{xy}$

$\left(2xy+\frac{1}{x}\right)dx + \left(x^2-\frac{1}{y}\right)dy$

$d(x^2y) + d\ln x - d\ln y = 0$

$x^2y + \ln x - \ln y = C$

$$M\,dx + N\,dy = 0$$
$$(M_1 + M_2)\,dx + (N_1+N_2)\,dy = 0$$
$$(M_1 + M_2)\,dx = 0 \quad | \cdot \mu_1(x,y)$$
$$(N_1 + N_2)\,dy = 0 \quad | \, \mu_2(x,y)$$

$$\mu = \mu_1 \Phi_1(U_1) = \mu_2 \Phi_2(U_2)$$

$$dU_1 = 0 \Rightarrow U_1$$
$$dU_2 = 0 \quad U_2$$

213. $y(x+y^2)dx + x^2(y-1)dy = 0$

$$xy\,dx - x^2 dy = 0 \quad | \cdot \frac{1}{x^2 y} = \mu_1$$
$$y^3 dx + x^2 y\,dy \quad | \cdot \frac{1}{x^2 y^3} = \mu_2$$

$$\frac{1}{x}dx - \frac{1}{y}dy = 0 \qquad \frac{1}{x^2}dx + \frac{1}{y^2}dy = 0$$
$$U = \frac{x}{y} \qquad U_2 = \frac{1}{x} + \frac{1}{y}$$

$g/3$ на 04.10 $\quad \Phi_1\left(\frac{x}{y}\right) = \frac{1}{y^2}\,\Phi_2\left(\frac{x+y}{xy}\right)$

$$F(x,y,y') = 0$$
$$y' = p\ ; \quad dy = p\,dx$$
$$F(x,y,p) = 0$$
$$y = f(x,p)$$
$$p\,dx = dy = f'_x\,dx + f'_p\,dx$$
$$x = g(p)$$
$$\begin{cases} y = f(g(p),p) \\ x = g(p) \end{cases}$$

269. $x = y'\sqrt{y'^2+1}$

$$y' = p$$
$$x = p\sqrt{p^2+1}$$

$$dx = (p^2+1)^{\frac{1}{2}} + \frac{p^2}{\sqrt{p^2+1}} = \frac{2p^2+1}{\sqrt{p^2+1}}$$

$$dx = \frac{dy}{p}$$

$$\int dy = \int p\,dx = \int \frac{1}{2}p \cdot \frac{2p^2+1}{\sqrt{p^2+1}}\,dp^2$$

$$y = \int \frac{2p^2}{\sqrt{p^2+1}}\,dp^2 + \frac{dp^2+1}{\sqrt{p^2+1}}$$

КР $xy' = y - xe^{\frac{y}{x}}$

$y' = \frac{y}{x} - e^{\frac{x}{y}}$

$\frac{dy}{dx} = \frac{xdt + tdx}{dx}$

$\frac{xdt + tdx}{dx} = t - e^{t}$

$xdt + tdx = tdx - e^{t}dx$

$\frac{-dt}{e^{t}} = \frac{dx}{x}$

$t = -\ln \ln cx$

$-\int \frac{dt}{e^{t}} = \int \frac{dx}{x}$

$\frac{y}{x} = -\ln \ln cx$

$e^{-t} = \ln x + \ln c$

$e^{-t} = \ln cx$

$-t = \ln \ln cx$

$y = -x \ln \ln Cx$

$2xy' - y = \ln y'$

$y' = p$

$2xp - y = \ln p$

$2[xdp + pdx] - dy = \frac{1}{p} dp$

$2[xdp + pdx] - pdx = \frac{1}{p} dp$

$2xdp + 2pdx - pdx$